Belacqua

Recirculating aquaculture plant for 100 beluga sturgeons in 20 tanks

System description, electrical design, control software and commissioning

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Status	Design example, not verified on a physical installation
Applies to	`belacqua_plc_001`, `belacqua_io_002`, `belacqua_io_003`, `belacqua_hmi_010`
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20	**4**	**369**	**146 · 46**
tanks, 100 fish, 1 t of biomass	MIAplc devices on one Ethernet network	ladder networks in 13 functions	alarms · trend recordings

IMPORTANT NOTICE

This document describes a design example for training and evaluation. It is not a construction document. MIAplc is not a safety controller and the example has not been tested on a plant: anyone who uses it as the basis for a real installation is responsible for its design, verification and compliance with the applicable regulations.

Revision history

Revision	Date	Description
1.0	5 November 2026	First issue.
1.1	13 November 2026	Editorial revision: document data and revision history, notice, scope, conventions and abbreviations added; sections, figures and tables numbered; no change to the design.

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1 About this document

1.1 Purpose and scope

This manual describes Belacqua, a complete design example for the MIAplc platform: a recirculating aquaculture plant for 100 beluga sturgeons in 20 rearing tanks, controlled by one PLC, two µLink I/O modules on the tank lines and a 10.1" touch panel. It shows how a plant of realistic size is structured with MIAplc, from the process data to the electrical design, the I/O lists, the PLC program, the operator panel and commissioning.

Belacqua is not an existing installation. The plant is sized on typical figures for recirculating systems, and the project files supplied with this manual have been checked with the MIAplc tools only (see section 12).

1.2 Intended audience

Automation engineers and technicians, teachers and students with a basic knowledge of PLCs and of electrical installations. Knowledge of fish farming is not required: section 2 gives the process data used in the design.

1.3 Conventions

> **WARNING**
> Marks information that concerns the safety of people or the protection of equipment.

> **NOTE**
> Marks additional information that helps to understand or to use the example.

Names of program objects, files, addresses and I/O points are written in `monospace`. Menu items of the MIAplc tools are written in *italics*, in the language of the tool.

1.4 Abbreviations

Table 1. Abbreviations used in this manual.

Abbreviation	Meaning
AI / AO	Analog input / analog output
DB	Data block of the PLC program
DI / DO	Digital input / digital output
FC	Function of the PLC program
HMI	Human-machine interface (operator panel)
I/O	Inputs and outputs

Abbreviation	Meaning
JB	Junction box on a tank line (JB-A, JB-B)
LD	Ladder diagram
MCP	Main control panel
PLC	Programmable logic controller
RAS	Recirculating aquaculture system
UPS	Uninterruptible power supply
UV	Ultraviolet steriliser
VFD	Variable-frequency drive
µLink	MIAplc protocol for remote I/O modules on Ethernet

1.5 Project files

The projects are in `MiaExamples/example_project/belacqua` and in the archive `belacqua-projects.zip` published with this manual.

Table 2. Project files of the example.

File	Tool	Content
`belacqua.mia`	MiaConfig	configuration with the four devices of the plant
`belacqua_plc_001.mia` · `.miaplc`	MiaConfig · PLC editor	hardware of the PLC board · PLC program
`belacqua_io_002.mia`, `belacqua_io_003.mia`	MiaConfig	hardware of the I/O modules of line A and line B
`belacqua_hmi_010.mia` · `.miahmi`	MiaConfig · panel editor	hardware of the panel · pages of the panel

2 Plant description

Beluga sturgeons (*Huso huso*) are bottom feeders that need cool, clean and well oxygenated water: 16–20 °C (optimum about 18 °C), dissolved oxygen above 7 mg/l, pH 7.0–8.0. Belacqua keeps 100 fish of about 10 kg, 5 per tank, in 20 round tanks of 3 m³ (Ø 2 m, 1 m of water). All the tanks share one recirculating water loop (RAS): the water leaves the tanks, is treated and returns; only a small quantity of new water is added every day.

Table 3. Process data and main equipment.

Item	Size	Design basis
Tanks	20 × 3 m³ = 60 m³	1 t of fish at about 17 kg/m³, a comfortable density for sturgeons
Circulation	60 m³/h at night, 90 m³/h by day, 120 m³/h boost	1 to 2 water exchanges per hour in every tank
Pumps	2 × 7.5 kW with VFD, duty / standby	one pump covers the full flow; the other covers faults and maintenance
Drum filter	120 m³/h, 60 µm screen, automatic backwash	removes faeces and uneaten feed
Biofilter	moving bed 12 m³, 2 blowers 2.2 kW with VFD	bacteria convert ammonia into nitrate
UV steriliser	2 × 400 W, 120 m³/h	limits bacteria and algae
Heat pump	12 kW heating and cooling	holds 18 °C in winter and in summer
Oxygen	liquid O2 tank, 2 oxygen cones, one valve per tank	pure oxygen where and when a tank needs it
Feed	about 10 kg/day (1 % of the weight), 6 meals	portion computed per tank from number of fish and weight

3 Hall layout

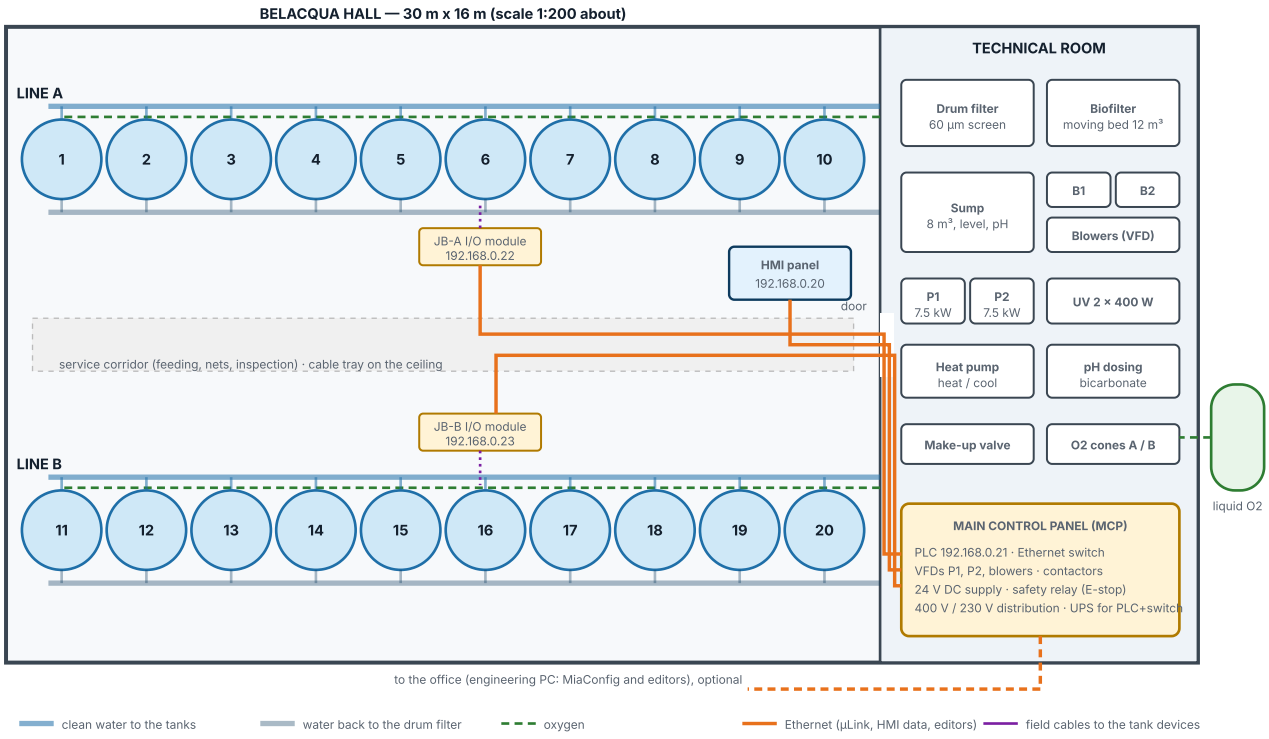


Figure 1. Hall of 30 × 16 m: two lines of 10 tanks with the service corridor between them, the technical room with the water treatment and the main control panel. Each line has a junction box (JB-A, JB-B) at its centre with the µLink I/O module: the field cables of the 10 tanks stay short, and only one Ethernet cable and the 24 V supply go back to the panel.

3.1 Location of the devices

- **Main control panel (MCP)**, technical room: PLC, Ethernet switch, VFDs, contactors, 24 V supply, safety relay, UPS.
- **JB-A, JB-B**: I/O module of the line, ADS1115 boards, input and output interface boards, terminals for the 10 tanks.
- **HMI panel**: at the end of the corridor, by the door of the technical room, with a view of all the tanks.

3.2 Equipment of each tank

- dissolved oxygen probe and temperature probe (4-20 mA);
- oxygen solenoid valve on the diffuser;
- automatic feeder with a hopper-empty sensor;
- overflow float and low-level float.

4 Network

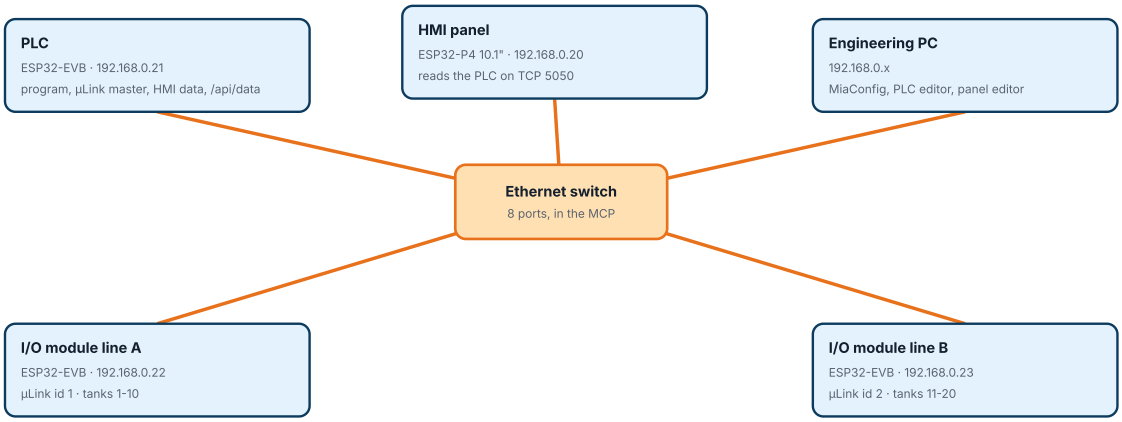


Figure 2. All the devices on one Ethernet switch (192.168.0.0/24). µLink carries the I/O of the line modules; the panel reads the PLC on TCP 5050; MiaConfig and the editors reach every board, also by MAC address when its IP address is unknown.

Table 4. Devices and addresses.

Device	Address	Project	Name on the network
PLC	192.168.0.21	`belacqua_plc_001`	`belacqua`
I/O module line A	192.168.0.22	`belacqua_io_002`	`linea` (µLink module and DB name)
I/O module line B	192.168.0.23	`belacqua_io_003`	`lineb`
HMI panel	192.168.0.20	`belacqua_hmi_010`	HMI-P4-xxxx

5 Electrical design

5.1 Safety concept

> **WARNING**
> MIAplc is not a safety controller. The emergency stop chain acts through a safety relay that opens the contactors and removes the VFD enables directly. The PLC reads `EStop_OK` only to stop its logic and to raise the alarm: no safety function depends on the program.

The drawings in this section are simplified and show the principle only. A real installation must be designed, built and verified by qualified personnel according to the regulations and standards that apply at the site, for example EN 60204-1 for the electrical equipment of machines and the IEC 61439 series for the control panel.

5.2 Main control panel

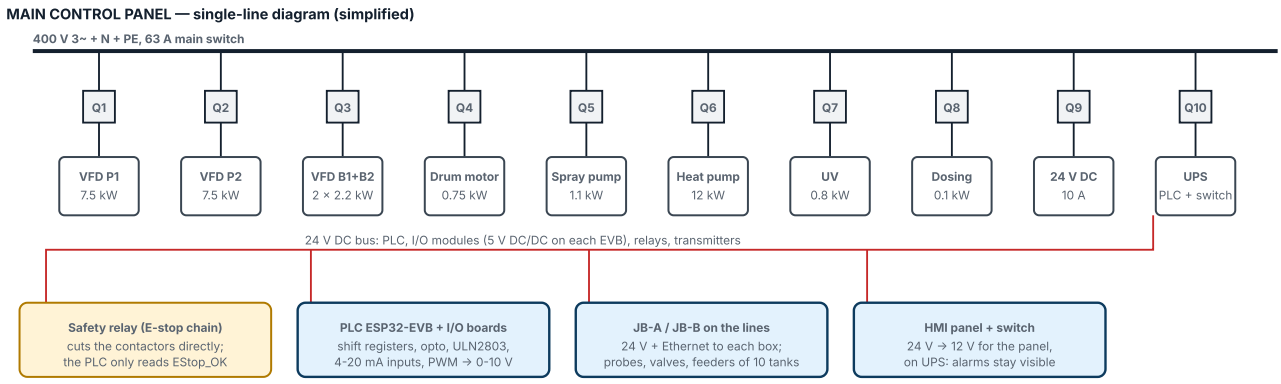


Figure 3. Single-line diagram of the main control panel. Motors on VFDs or contactors, all controlled by the PLC through 24 V interface relays; PLC and switch on a small UPS, so that alarms and data are kept during a power cut.

5.3 Controller interfaces

The ESP32-EVB boards are used with their Ethernet port only: the two relays of the board are not used. The digital I/O pass through shift registers on small interface boards, the analog inputs through 4-20 mA shunts.

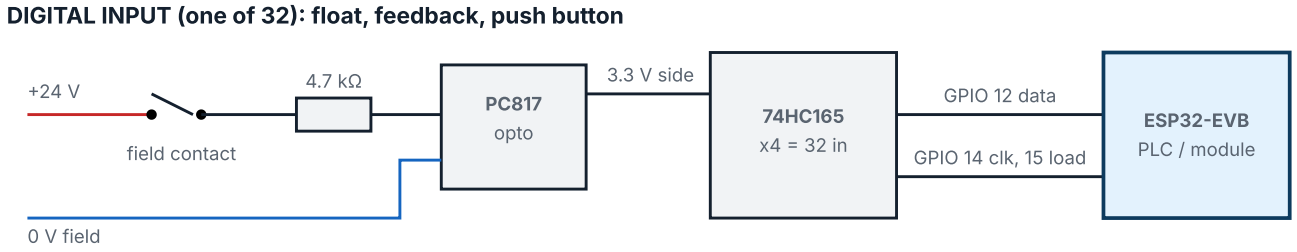


Figure 4. Digital input: 24 V field contact, optocoupler, shift register.

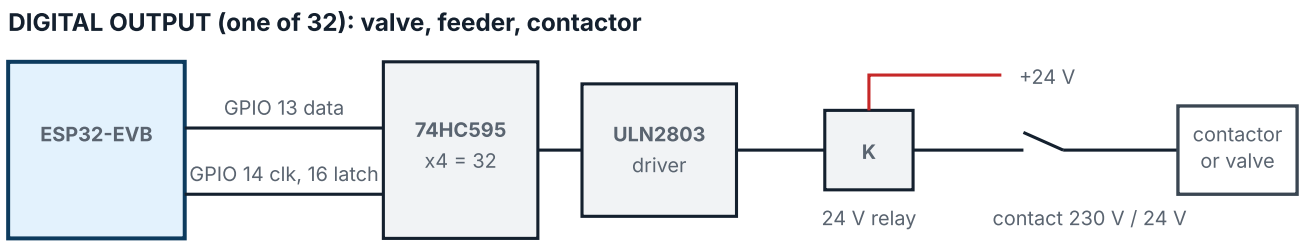


Figure 5. Digital output: shift register, driver, 24 V interface relay.

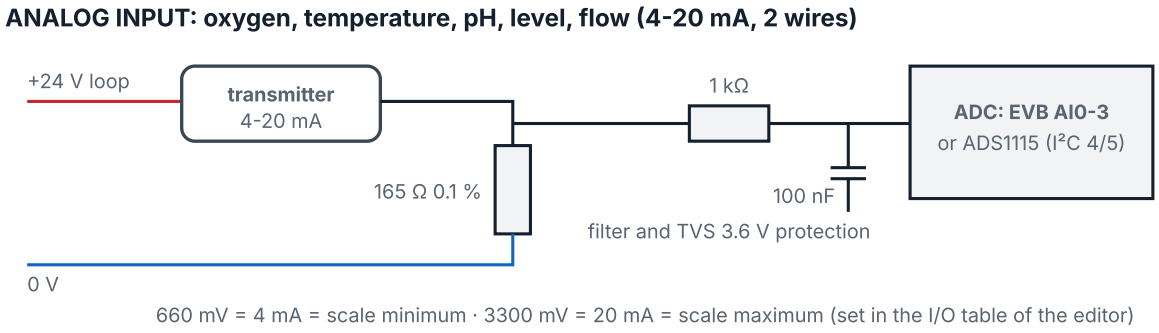


Figure 6. Analog input: 2-wire 4-20 mA transmitter on a precision shunt, with filter and protection.



Figure 7. Speed reference: PWM output converted to 0-10 V for the VFD.

Table 5. I/O capacity of the boards.

Board	Digital in	Digital out	Analog in	Analog out	Bus
PLC (MCP)	32 (4 x 74HC165)	32 (4 x 74HC595)	4 (EVB ADC)	4 PWM → 0-10 V	Ethernet LAN8720
I/O module line A, B (JB)	32	32	20 (4 EVB + 4 x ADS1115)	—	Ethernet, I²C SDA 4 / SCL 5

6 I/O lists

In the PLC program every point is a field of a DB generated from the I/O table: `HWDB` for the PLC board, `linea` and `lineb` for the modules. The analog values arrive already scaled (4-20 mA → engineering units) and averaged over 4 readings.

6.1 PLC (main control panel)

Table 6. I/O points of the PLC board.

Type	Point	Program name	Device	Range
DI	`sin0.0`	`HWDB.EStop_OK`	emergency stop chain closed (NC)	
DI	`sin0.1`	`HWDB.Pump1_Running`	pump P1 feedback	
DI	`sin0.2`	`HWDB.Pump2_Running`	pump P2 feedback	

Type	Point	Program name	Device	Range
DI	`sin0.3`	`HWDB.Pump1_Fault`	pump P1 VFD fault	
DI	`sin0.4`	`HWDB.Pump2_Fault`	pump P2 VFD fault	
DI	`sin0.5`	`HWDB.Drum_HighWater`	drum filter high water float	
DI	`sin0.6`	`HWDB.Sump_LowLow`	sump low-low float	
DI	`sin0.7`	`HWDB.Blower1_Running`	blower B1 feedback	
DI	`sin0.8`	`HWDB.Blower2_Running`	blower B2 feedback	
DI	`sin0.9`	`HWDB.UV_Running`	UV steriliser lamps on	
DI	`sin0.10`	`HWDB.Chiller_Fault`	chiller fault contact	
DI	`sin0.11`	`HWDB.Heater_Fault`	heater fault contact	
DI	`sin0.12`	`HWDB.Dosing_Tank_Low`	bicarbonate tank low level	
DI	`sin0.13`	`HWDB.Ack_Button`	alarm acknowledge push button	
DI	`sin0.14`	`HWDB.Power_OK`	mains power monitor	
DI	`sin0.15`	`HWDB.Drum_Running`	drum filter motor feedback	
DO	`sout0.0`	`HWDB.Pump1_Run`	pump P1 VFD run	
DO	`sout0.1`	`HWDB.Pump2_Run`	pump P2 VFD run	
DO	`sout0.2`	`HWDB.Drum_Motor`	drum filter motor	
DO	`sout0.3`	`HWDB.Drum_Spray`	drum filter backwash spray pump	
DO	`sout0.4`	`HWDB.Blower1_Run`	biofilter blower B1	
DO	`sout0.5`	`HWDB.Blower2_Run`	biofilter blower B2	
DO	`sout0.6`	`HWDB.UV_On`	UV steriliser	
DO	`sout0.7`	`HWDB.Makeup_Valve`	make-up water valve	
DO	`sout0.8`	`HWDB.Heater_On`	heater (heat pump) enable	
DO	`sout0.9`	`HWDB.Chiller_On`	chiller enable	
DO	`sout0.10`	`HWDB.Dosing_Pump`	pH dosing pump	
DO	`sout0.11`	`HWDB.Horn`	alarm horn	
DO	`sout0.12`	`HWDB.Beacon`	alarm beacon	
AI	`vpin.34`	`HWDB.Sump_Level`	sump level, ultrasonic 4-20 mA	0…200 cm
AI	`vpin.35`	`HWDB.Water_Temp`	loop temperature after the heat exchanger, PT100 4-20 mA	0…40 C
AI	`vpin.36`	`HWDB.Water_pH`	pH probe in the sump, 4-20 mA	0…14 pH

Type	Point	Program name	Device	Range
AI	`vpin.39`	`HWDB.Flow_Rate`	circulation flow, magnetic flow meter 4-20 mA	0…150 m3/h
AO	`pwm.2`	`HWDB.Pump_Speed`	speed reference to the pump VFDs (0-100 %)	0…100 %
AO	`pwm.4`	`HWDB.Blower_Speed`	speed reference to the blower VFDs (0-100 %)	0…100 %
AO	`pwm.5`	`HWDB.Dosing_Speed`	stroke rate of the pH dosing pump (0-100 %)	0…100 %

6.2 Line modules

The two modules are identical. Program names: `T01_O2`, `T01_Temp`, `T01_FeedEmpty`, `T01_Overflow`, `T01_LowLevel`, `T01_O2Valve`, `T01_Feeder` … `T20_…`; line signals `O2_Supply_OK`, `Aerator_Running`, `Light`, `Aerator`.

Table 7. I/O points of a line module (line A: tanks 1–10, line B: tanks 11–20).

Tank	Oxygen (AI)	Temperature (AI)	Feeder empty · overflow · low level (DI)	O2 valve · feeder (DO)
tank 1 (A) / 11 (B)	`vpin.34`	`xadc1.2`	`sin0.0` · `sin0.1` · `sin0.2`	`sout0.0` · `sout0.1`
tank 2 (A) / 12 (B)	`vpin.35`	`xadc1.3`	`sin0.3` · `sin0.4` · `sin0.5`	`sout0.2` · `sout0.3`
tank 3 (A) / 13 (B)	`vpin.36`	`xadc2.0`	`sin0.6` · `sin0.7` · `sin0.8`	`sout0.4` · `sout0.5`
tank 4 (A) / 14 (B)	`vpin.39`	`xadc2.1`	`sin0.9` · `sin0.10` · `sin0.11`	`sout0.6` · `sout0.7`
tank 5 (A) / 15 (B)	`xadc0.0`	`xadc2.2`	`sin0.12` · `sin0.13` · `sin0.14`	`sout0.8` · `sout0.9`
tank 6 (A) / 16 (B)	`xadc0.1`	`xadc2.3`	`sin0.15` · `sin0.16` · `sin0.17`	`sout0.10` · `sout0.11`
tank 7 (A) / 17 (B)	`xadc0.2`	`xadc3.0`	`sin0.18` · `sin0.19` · `sin0.20`	`sout0.12` · `sout0.13`
tank 8 (A) / 18 (B)	`xadc0.3`	`xadc3.1`	`sin0.21` · `sin0.22` · `sin0.23`	`sout0.14` · `sout0.15`
tank 9 (A) / 19 (B)	`xadc1.0`	`xadc3.2`	`sin0.24` · `sin0.25` · `sin0.26`	`sout0.16` · `sout0.17`
tank 10 (A) / 20 (B)	`xadc1.1`	`xadc3.3`	`sin0.27` · `sin0.28` · `sin0.29`	`sout0.18` · `sout0.19`
line	—	—	`sin0.30` O2 supply OK · `sin0.31` aerator running	`sout0.20` lights · `sout0.21` emergency aerator

7 PLC program

7.1 Structure

Ladder (LD) in 13 functions called by Main: 369 networks, about 30 KB of program, at most 3300 instructions per scan. Data in 6 DBs: `Settings` and `TankSettings` (retentive: they are kept through a power cut), `Process`, `Tanks`, `Alarms`, `Stats` (retentive).

Table 8. Functions of the PLC program.

FC	Function	Description
1	Clock and schedules	Local time from the PLC clock, day of the week, day / night, peak tariff, counters of the day.
2	Water level	Make-up valve with hysteresis; low-low level stops pumps and heating.
3	Recirculation	Flow target by day, at night (lower) and with a water alert (higher); the pump speed follows the flow in 5 s steps, so the pumps run at the lowest speed that gives it. Duty/standby pumps swap every 24 h and on a fault.
4	Drum filter	Backwash on high water, every N minutes and every night at a set time, when energy costs less.
5	Biofilter and UV	Blower 1 always, blower 2 by day or with a water alert, blower speed day / night; UV only with flow.
6	Oxygen	Per tank: oxygen valve with hysteresis around the set point; low and critical oxygen per line; emergency aerator of the line while a tank is critical.
7	pH dosing	Bicarbonate in pulses (on / pause) only with flow; timeout alarm if the pH does not rise.
8	Temperature	Heat pump in heating or cooling with a band; wider band in the peak tariff hours; never both together.
9	Feeding	Up to 8 meals a day, a fasting day, portion per tank = fish x weight x daily % / meals; a tank with low oxygen or bad temperature skips the meal; tank lights per line on the chosen days.
10	Tank alarms	An alarm word per tank: O2 low, O2 critical, feeder empty, overflow, low level, temperature.
11	Plant alarms	26 latched alarms; acknowledge from the panel or the push button; horn until acknowledged, beacon.
12	Tank settings	The panel edits one tank at a time (in service, modes, fish, weight) and writes it back.
13	Statistics	Running hours, backwashes, meals, feed given and estimated pump energy of the day.

7.2 Operating modes and interlocks

Every actuator has a mode: **Auto** (the program decides), **Manual ON**, **Manual OFF**.

> **NOTE**
> Interlocks stay active in manual mode: no pump without water level, no UV or heating without flow, no feeder with an empty hopper.

7.3 Energy saving

The power drawn by a centrifugal pump varies with the cube of its speed: at 70 % speed it is about 34 % of the rated power. Running the pumps at the lowest speed that gives the required flow, and at 60 m³/h instead of 90 m³/h for the 11 hours of night, is estimated to save about 40 % of the daily pumping energy. The blowers slow down at night and the second one stops; heating and cooling use a wider band in the peak tariff hours; the drum filter is washed at 03:00.

7.4 Data for other systems

With **data exposure** the PLC publishes oxygen, temperature and alarms of every tank and the plant values as JSON (`http://192.168.0.21/api/data`) and accepts four set points: a farm management software can read the plant without changes to the program.

8 HMI panel

8.1 Design principles and page structure

The panel follows the high-performance HMI principles of ISA-101: neutral grey screens; colour only for abnormal conditions (amber for a warning, red for an alarm, blinking for a critical alarm); analog values as a pointer on a scale with the normal band in darker grey; equipment drawn filled when running and hollow when stopped; the same header, alarm line and navigation on every page. The pages are organised in three levels.

Table 9. Pages of the panel.

Level	Page	Content
1	Overview	the hall: the 20 tanks in their position, each with oxygen, temperature, O2 and FEED marks, amber or red when the tank has a warning or an alarm; the technical room with flow, temperature, pH, level, pumps and blowers
2	Water treatment	the recirculation loop drawn as a flow diagram, values, modes of pumps, drum filter, blowers, UV, make-up
2	Water quality	oxygen of all 20 tanks as bars on the normal band, oxygen, pH and temperature set points and modes
2	Feeding · Energy · Trends · Alarms · Tank settings · Statistics	meal times, schedules and energy saving, 6-hour trends, alarm list and history, settings of one tank at a time, running hours and totals of the day
3	Tank 1 … Tank 20	one page per tank: the tank drawn with feeder, oxygen diffuser and valve, the most severe alarm in words, oxygen and temperature indicators, the 6-hour trend of the tank, previous / next tank

8.2 Screens

> **NOTE**
> The pictures are taken from the panel editor with sample values, without field devices.

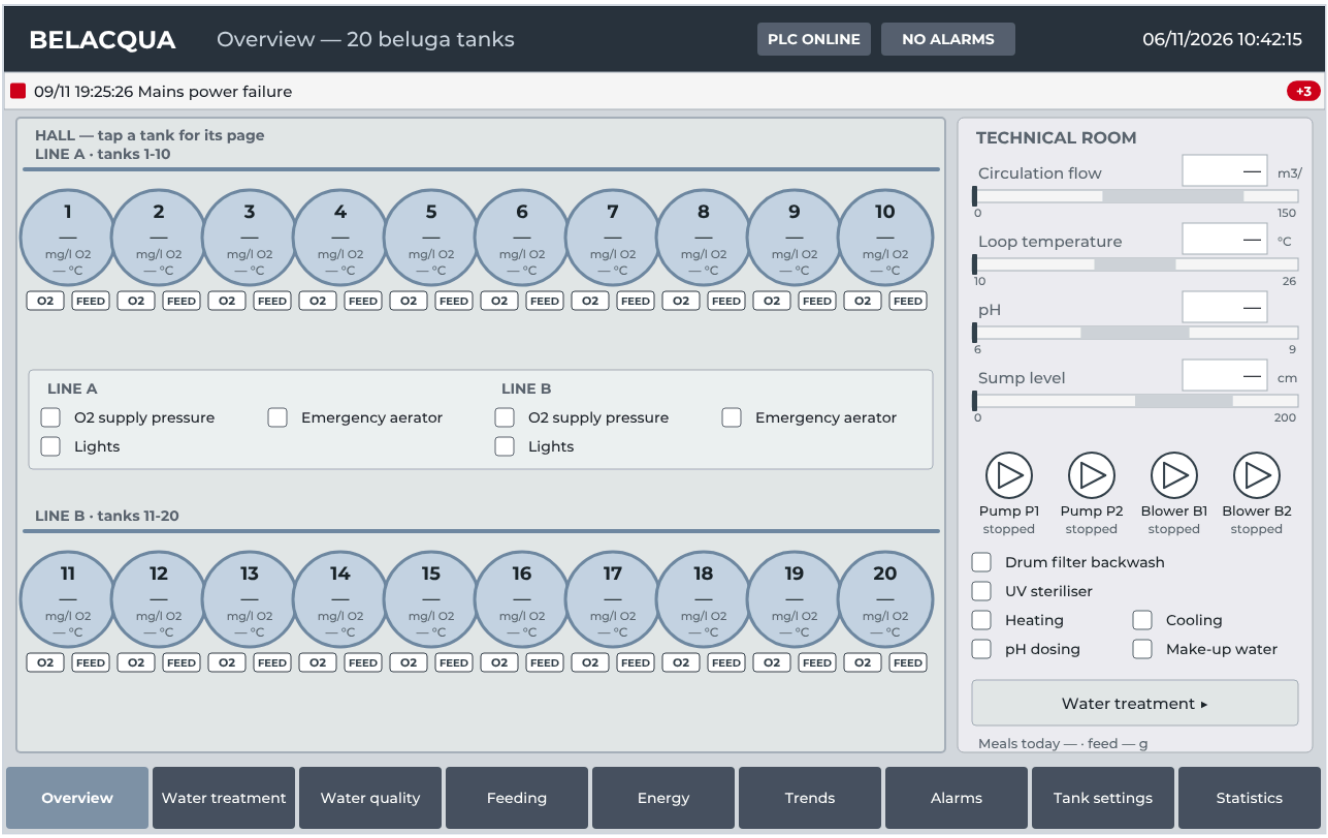


Figure 8. Level 1, overview: the 20 tanks in their position in the hall. Grey-blue is normal; tanks 4 and 12 have a warning (amber), tank 9 an alarm (red), tank 16 a critical oxygen alarm (red, blinking: captured here in the off phase). O2 and FEED under a tank: its oxygen valve or feeder is on. Right: the technical room with flow, loop temperature, pH and level on scales with the normal band, pumps and blowers filled when running.

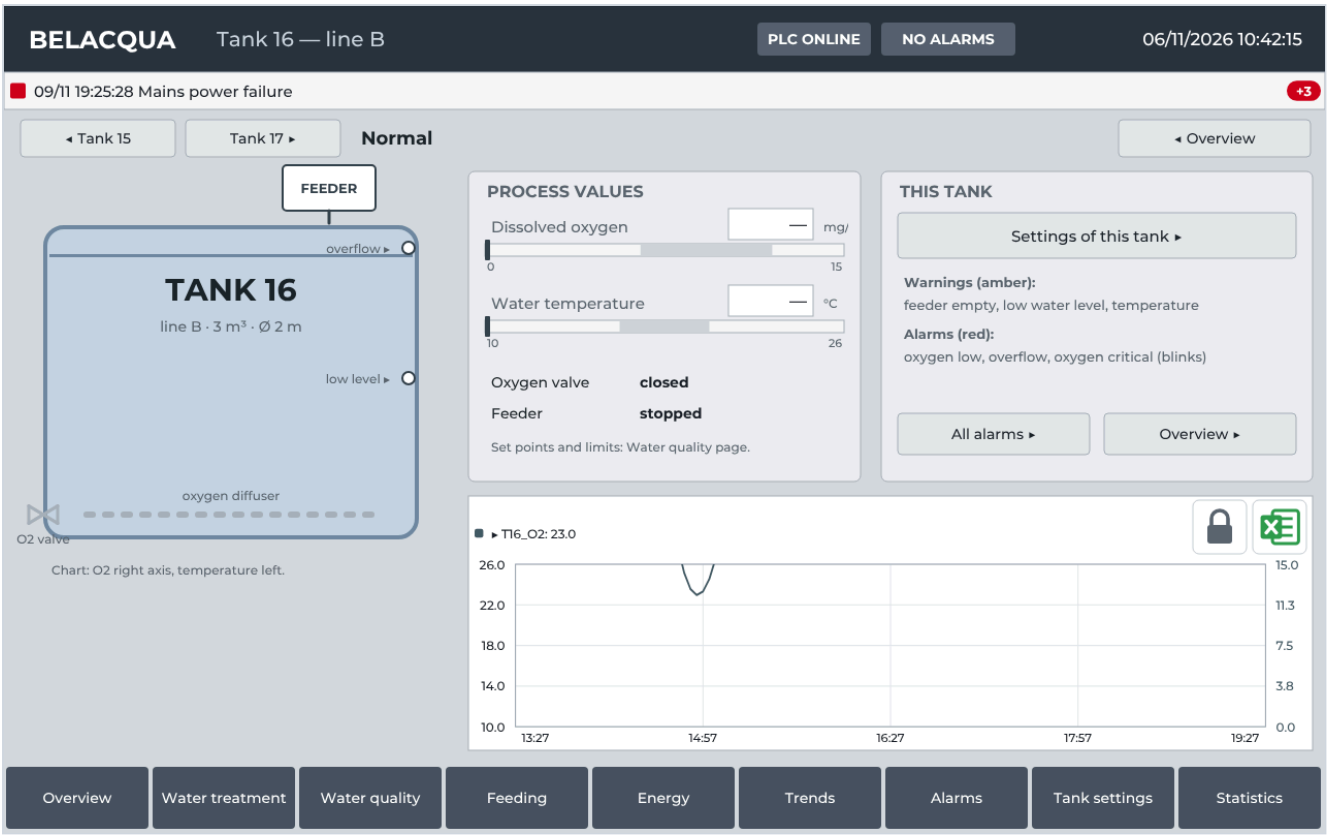


Figure 9. Level 3, page of a tank: the most severe alarm in words, the tank drawn with feeder, oxygen diffuser and valve, oxygen and temperature on their scales (the value turns amber outside the normal band), the 6-hour trend of the tank (empty in this picture: no data recorded).

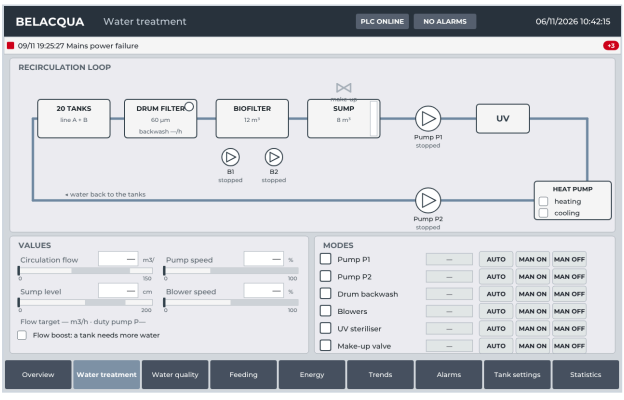


Figure 10. Level 2, water treatment: the recirculation loop, values and the modes of the machines (AUTO, MAN ON, MAN OFF; amber when not in automatic).

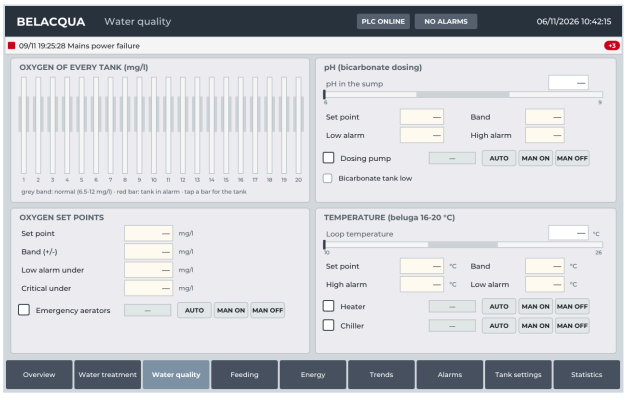


Figure 11. Level 2, water quality: the oxygen of all 20 tanks on the normal band, set points of oxygen, pH and temperature.

The panel exchanges data with the PLC through the **HMI data** table of the PLC program (240 of 256 rows: 4 per tank plus the plant). It requires the panel firmware 0.1.8 or later (64 recordings, 20 series per chart).

9 Alarms

26 plant alarms, latched in the PLC until acknowledged, plus 6 alarms per tank (feeder empty, low water level, temperature out of range, O2 low, overflow, O2 critical) from the alarm word of the tank: 146 alarms on the panel. Red marks an alarm, which requires immediate action; amber marks a warning.

Table 10. Plant alarms.

No.	Plant alarm	PLC field
1	Emergency stop pressed	`Al_EStop`
2	Mains power failure	`Al_Power`
3	Pump P1 fault	`Al_Pump1`
4	Pump P2 fault	`Al_Pump2`
5	No circulation flow	`Al_No_Flow`
6	Sump level low-low: pumps stopped	`Al_Level_LowLow`
7	Drum filter screen clogged	`Al_Drum_Clogged`
8	Drum filter motor fault	`Al_Drum_Motor`
9	Biofilter blower fault	`Al_Blowers`
10	UV steriliser fault	`Al_UV_Fault`
11	UV lamps to replace	`Al_UV_Service`
12	Oxygen supply pressure low, line A	`Al_O2_Supply_A`
13	Oxygen supply pressure low, line B	`Al_O2_Supply_B`
14	Emergency aerator line A not running	`Al_Aerator_A`
15	Emergency aerator line B not running	`Al_Aerator_B`
16	pH too low	`Al_pH_Low`
17	pH too high	`Al_pH_High`
18	pH dosing too long: check the bicarbonate tank	`Al_pH_Dose_Timeout`
19	Bicarbonate tank low	`Al_Dosing_Tank_Low`
20	Water temperature high	`Al_Temp_High`
21	Water temperature low	`Al_Temp_Low`
22	Chiller fault	`Al_Chiller`
23	Heater fault	`Al_Heater`
24	I/O module line A (uLink) not answering	`Al_Module_A`
25	I/O module line B (uLink) not answering	`Al_Module_B`

No.	Plant alarm	PLC field
26	PLC clock not set: feeding and schedules stopped	`Al_Clock`

10 Settings

All the settings are in the retentive DB `Settings` and can be changed on the panel within limits.

Table 11. Settings and default values.

Name	Type	Default	Meaning
`Level_Makeup_On`	REAL	120.0	sump level (cm) under which the make-up valve opens
`Level_Makeup_Off`	REAL	130.0	sump level (cm) at which the make-up valve closes
`Level_LowLow`	REAL	80.0	sump level (cm) under which pumps and heating stop
`Flow_Day`	REAL	90.0	circulation flow by day (m3/h): 20 tanks x 3 m3 × 1.5 exchanges per hour
`Flow_Night`	REAL	60.0	circulation flow at night (m3/h): energy saving, fish at rest
`Flow_Boost`	REAL	120.0	circulation flow when the water is not good (low oxygen, temperature)
`Flow_Band`	REAL	3.0	flow band (m3/h): the speed changes only outside it
`Speed_Step`	REAL	1.0	pump speed change every 5 s (%)
`Speed_Min`	REAL	35.0	minimum pump speed (%)
`Speed_Max`	REAL	100.0	maximum pump speed (%)
`Flow_Min`	REAL	30.0	flow (m3/h) under which there is no circulation
`Pump_Swap_s`	UDINT	86400	running time (s) after which the duty pump changes
`Blower_Day`	REAL	85.0	blower speed by day (%)
`Blower_Night`	REAL	60.0	blower speed at night (%): one blower only
`Drum_Interval`	TIME	1800000	drum filter backwash every ... (ms) when the float does not ask before
`Drum_Wash`	TIME	30000	duration of one backwash (ms)
`Drum_Max_Hour`	INT	8	backwashes in one hour above which the screen is clogged
`Drum_Night_Wash`	UDINT	180	scheduled backwash at ... (minute of the day, 180 = 03:00, cheap energy)
`O2_Set`	REAL	8.0	dissolved oxygen set point (mg/l)
`O2_Band`	REAL	0.5	oxygen valve band (mg/l): on under Set-Band, off over Set+Band
`O2_Low`	REAL	6.5	low oxygen in a tank (mg/l): alarm, no feeding in that tank, flow boost
`O2_Critical`	REAL	5.0	critical oxygen (mg/l): emergency aerator of the line on
`pH_Set`	REAL	7.4	pH set point
`pH_Band`	REAL	0.1	pH dosing band: dosing under Set-Band until Set+Band

Name	Type	Default	Meaning
`pH_Low`	REAL	6.8	low pH alarm
`pH_High`	REAL	8.2	high pH alarm
`pH_Dose_On`	TIME	2000	dosing pulse (ms)
`pH_Dose_Pause`	TIME	20000	pause between two pulses (ms): mixing time
`pH_Dose_Max`	TIME	900000	pH still low after dosing for ... (ms): alarm
`Dosing_Speed_Set`	REAL	50.0	stroke rate of the dosing pump (%)
`Temp_Set`	REAL	18.0	water temperature set point (C), beluga: 16-20 C
`Temp_Band`	REAL	0.5	temperature band off-peak (C)
`Temp_Band_Peak`	REAL	1.5	temperature band in the peak tariff hours (C): energy saving
`Temp_High`	REAL	21.0	high temperature alarm (C)
`Temp_Low`	REAL	14.0	low temperature alarm (C)
`Peak_Start`	UDINT	480	peak tariff from ... (minute of the day, Monday-Friday)
`Peak_End`	UDINT	1140	peak tariff until ... (minute of the day)
`Day_Start`	UDINT	420	day from ... (minute of the day): lights, day flow
`Day_End`	UDINT	1200	night from ... (minute of the day)
`Light_Days`	INT	127	lights on these days: bit 0 Monday ... bit 6 Sunday (127 = every day)
`Feed_Enable`	BOOL	True	automatic feeding enabled
`Fast_Day`	INT	6	fasting day, no feeding: 0 Monday ... 6 Sunday, 7 = none
`Feed_Pct`	REAL	1.0	daily feed in % of the fish weight
`Feeder_Rate`	REAL	5.0	feeder output (g/s), the same model on every tank
`Meal_1`	UDINT	480	meal 1: minute of the day (480 = 08:00), 9999 = off
`Meal_2`	UDINT	600	meal 2: minute of the day (480 = 08:00), 9999 = off
`Meal_3`	UDINT	720	meal 3: minute of the day (480 = 08:00), 9999 = off
`Meal_4`	UDINT	840	meal 4: minute of the day (480 = 08:00), 9999 = off
`Meal_5`	UDINT	960	meal 5: minute of the day (480 = 08:00), 9999 = off
`Meal_6`	UDINT	1080	meal 6: minute of the day (480 = 08:00), 9999 = off
`Meal_7`	UDINT	9999	meal 7: minute of the day (480 = 08:00), 9999 = off
`Meal_8`	UDINT	9999	meal 8: minute of the day (480 = 08:00), 9999 = off
`UV_Service_h`	REAL	9000.0	UV lamp hours before replacement
`Time_Offset_s`	UDINT	3600	local time offset from UTC (s): 3600 winter, 7200 summer (Italy)

11 Build and commissioning

11.1 Engineering tools

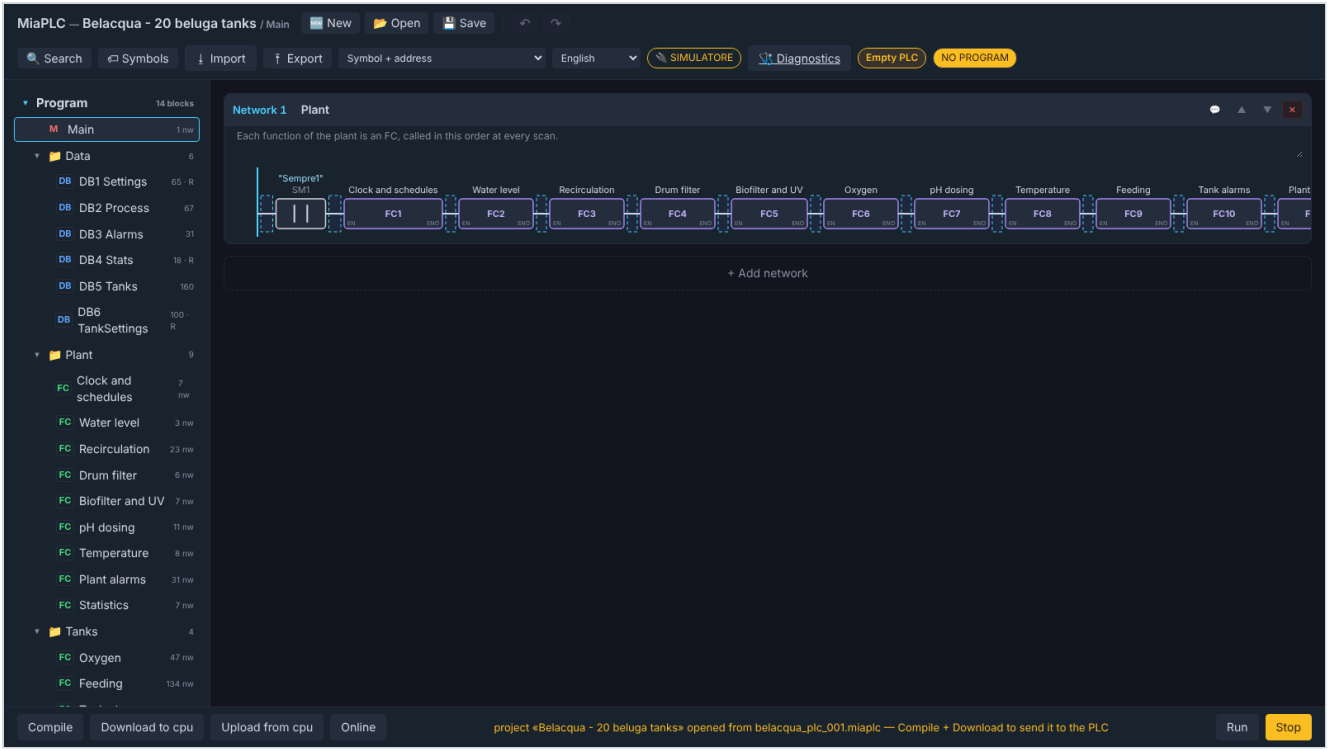


Figure 12. The MIAplc editor (in the browser, served by the PLC itself) with the Belacqua program: on the left the blocks in folders (data, plant, tanks, system DBs of the board and of the two µLink modules), in the centre Main calling the 13 functions.

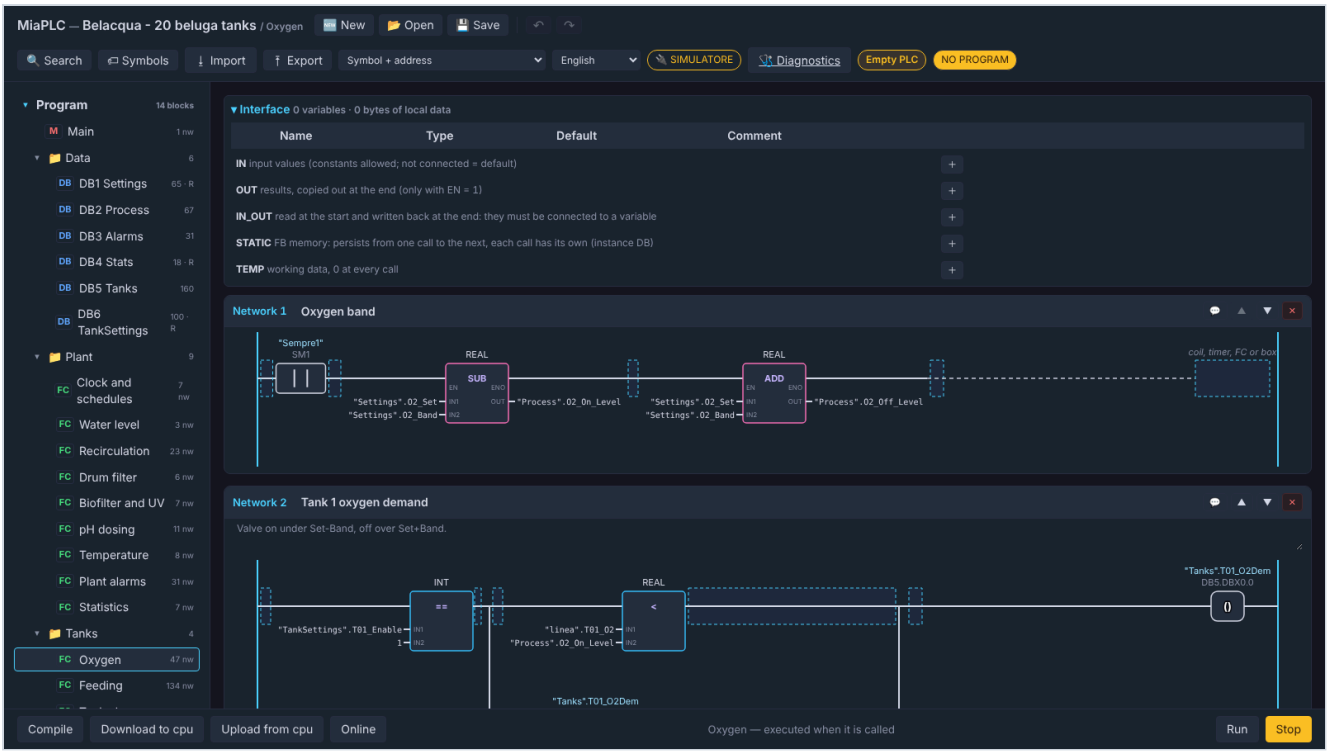


Figure 13. The ladder of FC6 Oxygen: the band around the set point, then for each tank the oxygen demand with hysteresis and the valve.

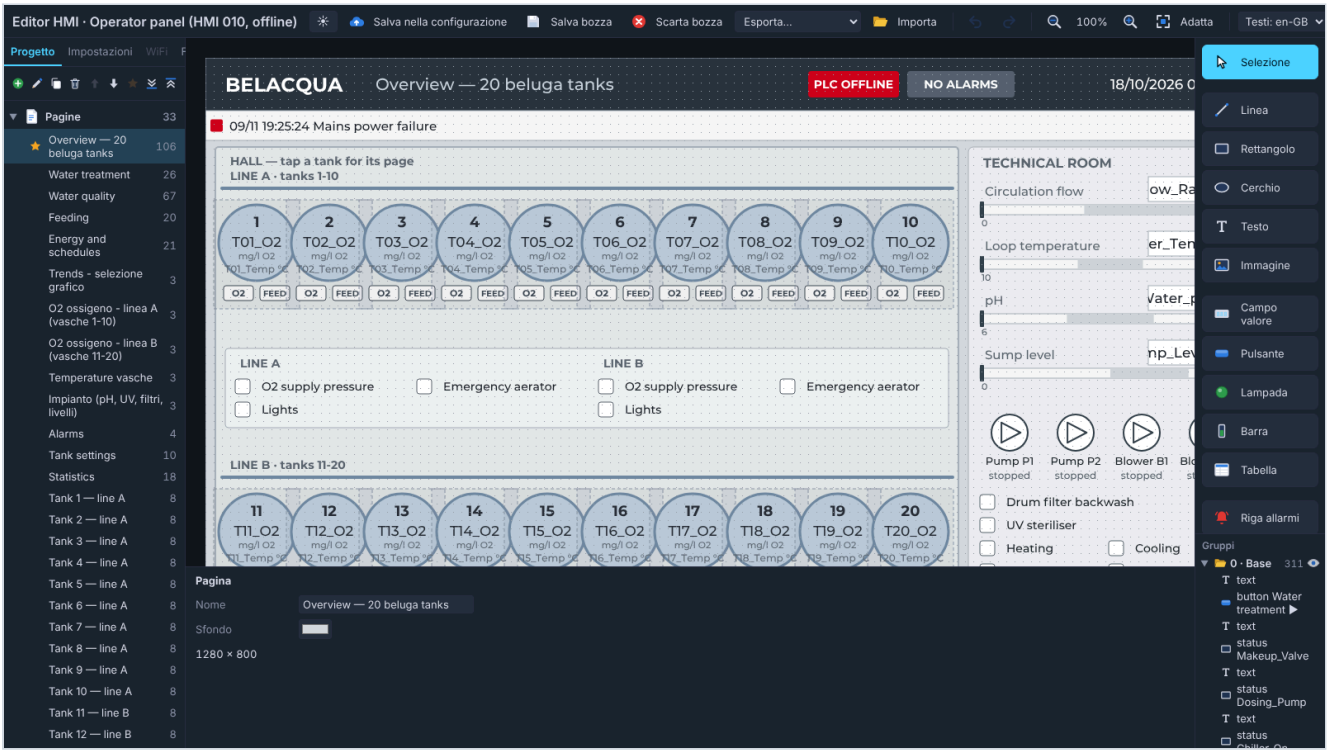


Figure 14. The panel editor (in the browser, served by the panel): the 29 pages on the left, the page in the centre, the objects on the right. *Importa* loads `belacqua_hmi_010.miahmi`; *Pubblica sul pannello* makes it the project of the panel.

11.2 Procedure

1. **Firmware.** Open each `.mia` in MiaConfig, *Genera → Compila firmware*, and load it on the board by USB (first time) or over the network. The PLC project uses developer mode (required for the HMI data).
2. **Network.** Every board starts with the address of its project; a board that cannot be reached can be found and set by MAC address in MiaConfig (*Schede in rete*).
3. **Modules.** On the PLC, Diagnostics → I/O modules: pair `linea` and `lineb`.
4. **PLC program.** Open `belacqua_plc_001.miaplc` in the PLC editor, connect to 192.168.0.21, check the I/O table against the real board, Compile, Download with the PLC in STOP, then RUN.
5. **Panel.** In the panel editor (http://192.168.0.20) *Importa* `belacqua_hmi_010.miahmi` and publish; in the panel settings set the PLC address 192.168.0.21.
6. **Clock.** Set the PLC date and time from the panel (Energy page): schedules and feeding depend on it.
7. **Checks.** Each tank in Manual ON / OFF for valve and feeder, probes against a hand-held meter, every alarm once, emergency stop, power cut with the UPS.

12 Verification status

The example has been verified with the MIAplc tools. It has not been loaded on devices or tested on a plant.

Table 12. Verification carried out.

Part	Verification with the tools	On real devices
MiaConfig projects (4)	MiaConfig checks: no errors	not performed
PLC program	editor compiler: no errors, no warnings	not performed
HMI panel project	panel project checks (firmware 0.1.8): valid	not performed
µLink module numbers	the program reads `ulink_ok` bit 1 (line A) and bit 2 (line B)	to be checked
Electrical drawings	example design, to be adapted to the real plant and to the applicable regulations	not applicable

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